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Worksheet Problems on Data Structures and Algorithms

Worksheet 8

Q1: Roman Number to Integer

Given a string in roman no format (s) your task is to convert it to an integer . Various symbols and their values are given below.
I 1
V 5
X 10
L 50
C 100
D 500
M 1000

Example 1:

Input:

s = V

Output: 5

Example 2:

Input:

s = III

Output: 3

Q2: Anagram

Given two strings s1 and s2 consisting of lowercase characters. The task is to check whether two given strings are an anagram of each other or not. An anagram of a string is another string that contains the same characters, only the order of characters can be different. For example, act and tac are an anagram of each other. Strings s1 and s2 can only contain lowercase alphabets.

Note: You can assume both the strings s1 & s2 are non-empty.

Examples :

Input: s1 = "geeks", s2 = "kseeg"

Output: true

Explanation: Both the string have same characters with same frequency. So, they are anagrams.

Input: s1 = "allergy", s2 = "allergic"

Output: false

Explanation: Characters in both the strings are not same, so they are not anagrams.

Input: s1 = "g", s2 = "g"

Output: true

Explanation: Character in both the strings are same, so they are anagrams.

Q3: Remove Duplicates

Given a string s without spaces, the task is to remove all duplicate characters from it, keeping only the first occurrence.

Note: The original order of characters must be kept the same.

Examples :

Input: s = "zvvo"

Output: "zvo"

Explanation: Only keep the first occurrence

Input: s = "gfg"

Output: "gf"

Explanation: Only keep the first occurrence

Q4: Implement Atoi

Given a string s, the objective is to convert it into integer format without utilizing any built-in functions. Refer the below steps to know about atoi() function.

Cases for atoi() conversion:

1. **Skip any leading whitespaces.**
2. **Check for a sign (‘+’ or ‘-‘), default to positive if no sign is present.**
3. **Read the integer by ignoring leading zeros until a non-digit character is encountered or end of the string is reached. If no digits are present, return 0.**
4. **If the integer is greater than 231 – 1, then return 231 – 1 and if the integer is smaller than -231, then return -231.**

Examples:

Input: s = "-123"

Output: -123

Explanation: It is possible to convert -123 into an integer so we returned in the form of an integer

Input: s = " -"

Output: 0

Explanation: No digits are present, therefore the returned answer is 0.

Input: s = " 1231231231311133"

Output: 2147483647

Explanation: The converted number will be greater than 231 – 1, therefore print 231 – 1 = 2147483647.

Input: s = "-999999999999"

Output: -2147483648

Explanation: The converted number is smaller than -231, therefore print -231 = -2147483648.

Input: s = " -0012gfg4"

Output: -12

Explanation: Nothing is read after -12 as a non-digit character 'g' was encountered.

Q5: Longest Common Prefix of Strings

Given an array of strings arr. Return the longest common prefix among each and every strings present in the array. If there's no prefix common in all the strings, return "-1".

Examples :

Input: arr[] = ["geeksforgeeks", "geeks", "geek", "geezer"]

Output: gee

Explanation: "gee" is the longest common prefix in all the given strings.

Input: arr[] = ["hello", "world"]

Output: -1

Explanation: There's no common prefix in the given strings.